

LLM-Based Detection and Revision of Questionnaire Items

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Abstract

Questionnaire defects can affect how respondents understand a question, retrieve information, form a judgement, and map that judgement onto the available response options. Large language models (LLMs) could support questionnaire review, but useful automated appraisal requires more than identifying isolated flaws: a system should also preserve acceptable items, recover multiple independently supported defects, and revise problematic items without changing the intended construct. This study evaluates locally deployed large language models on an expert-reviewed benchmark of 200 questionnaire items covering 16 defect categories, including clean, single-defect, and multi-defect cases, with evaluation of both defect detection and construct-preserving revision. A simple checker–reviser pipeline is compared with a role-separated orchestrated workflow under three prompt conditions: a zero-shot taxonomy codebook, additional operational rules, and family-spanning few-shot demonstrations. Performance varied substantially across models, prompting strategies, and architectures. Qwen achieved the strongest overall results, with P2 orchestration reaching a micro-F1 of 0.563, an exact label-set match of 0.430, and a clean-item false-positive rate of 0.050. The baseline generally achieved higher recall and broader multi-label coverage, but also produced more unsupported labels, whereas orchestration was more selective and preserved clean items more reliably. Few-shot calibration improved selected response-option and scale decisions, but gains were uneven across models and defect categories. Thinking mode improved detection among successfully completed outputs. Overall, the results show that LLM-based questionnaire appraisal involves a trade-off between defect coverage, restraint, revision quality, and operational reliability, and that structured prompting and orchestration can improve selected aspects of performance without yet providing consistently reliable end-to-end automation. The repository can be accessed at: <https://github.com/sn1887/SEMINAR-ITEM-REVISER>.

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1 Introduction

Poorly designed survey questions can distort the data they are intended to collect. Problems may arise from the wording of the question itself or from the response options through which respondents are asked to express an answer. Classic questionnaire-design examples illustrate how easily such problems can arise. Saris and Gallhofer, for instance, discuss an item that asks respondents to evaluate the European Parliament and the European Commission using a single response. Because a respondent may judge the two institutions differently, the item combines distinct evaluations and is therefore double-barreled. They also discuss the seemingly straightforward question “What is the best book you read last year?”, which presupposes that the respondent read at least one book during the specified period [9]. Such flaws are easy to introduce but can prevent respondents from answering as intended. Even minor design choices can shape how respondents interpret a question and formulate an answer; when the cognitive burden becomes too high, they may satisfice rather than engage carefully with the task [4, 11].

Established approaches for identifying these problems include expert review, questionnaire appraisal, cognitive interviewing, behaviour coding, respondent debriefing, and field pretesting [7]. These procedures provide different kinds of evidence and remain important for questionnaire development, but they also require methodological expertise, appropriate participants, and iterative review. An automated system could therefore be useful as an early screening and revision aid, particularly when specialist survey-methods expertise is not immediately available. Such assistance should complement rather than replace expert and respondent-based evaluation, since problems that emerge during actual interpretation and response may not be visible from the questionnaire text alone.

Against this background, we investigate whether locally deployed LLMs can support questionnaire review by distinguishing acceptable items from problematic ones, identifying the specific design problems present, and proposing targeted revisions without changing what the question is intended to measure. The evaluation uses an expert-reviewed benchmark of 200 questionnaire items under a closed 16-label taxonomy and evaluates both defect detection and construct-preserving revision. Two ways of organizing the same underlying LLM are compared: a simple checker–reviser workflow and a role-separated workflow that divides detection, planning, revision, and validation across specialized calls. The study further examines whether performance changes when the models receive increasingly explicit guidance, moving from taxonomy definitions to operational rules and then to few-shot examples.

The study is guided by three research questions:

Research Questions

- RQ1:** How accurately can locally deployed LLMs distinguish acceptable questionnaire items from items containing one or more taxonomy-defined defects?
- RQ2:** How does a role-separated workflow compare with a simpler checker–reviser workflow in defect detection, clean-item preservation, and construct-preserving revision?
- RQ3:** How does increasingly explicit prompt guidance affect detection and revision performance, and are these effects consistent across defect families?

2 Related Work

Prior work relevant to this study can be organized around two related questions: how questionnaire problems can be identified before data collection, and how recent LLM-based methods extend such appraisal toward automated critique and revision. The first line of work establishes explicit principles

and tools for questionnaire evaluation, while the second examines how reliably LLMs can apply related judgments without continuous expert intervention.

2.1 Structured and Computer-Assisted Questionnaire Appraisal

Questionnaire evaluation methods differ in both the problems they target and the evidence on which their judgments are based. Expert appraisal evaluates an instrument against known design principles, whereas cognitive interviewing, behaviour coding, and field pretesting provide progressively more direct evidence about how respondents interact with a question [7]. Saris and Gallhofer provide a broader framework for questionnaire design in which the measured concept, formulation of the request, response alternatives, scale properties, and overall survey-item structure are treated as consequential design choices [9]. Their discussion covers problems such as double-barreled requests, implicit assumptions, complex formulations, incomplete or overlapping response alternatives, mismatches between the request and its response categories, and choices in scale construction. These principles are relevant to the present study beyond their later implementation in SQP and provide methodological grounding for several defect mechanisms examined in the benchmark.

Structured and computer-assisted approaches operationalize these principles in different ways. The Question Appraisal System (QAS-99) organizes expert appraisal into explicit decisions concerning potential questionnaire problems [12]. QUAID instead provides automated screening for selected comprehension risks, including unfamiliar or vague terms, ambiguous noun phrases, complex syntax, and working-memory demands [3]. The Survey Quality Predictor (SQP) addresses a different target: it uses coded characteristics of survey questions and evidence from multitrait-multimethod experiments to predict reliability, validity, and overall measurement quality [9]. These systems therefore range from structured diagnosis of potential problems to automated comprehension screening and prediction of expected measurement quality.

The methods should not be treated as interchangeable. Maitland and Presser compared QUAID, SQP, expert review, QAS, and cognitive interviewing against problems observed under more typical survey conditions and found that the approaches identified partly different problems and differed between subjective and objective questions [5]. Their results support the use of multiple forms of evidence rather than treating any single appraisal procedure as sufficient. For the present study, this distinction motivates viewing automated LLM appraisal as an early screening and revision layer whose outputs remain subject to methodological evaluation rather than as a replacement for human or respondent-based testing.

2.2 LLMs for Questionnaire Design, Pretesting, and Appraisal

Recent LLM research has first examined whether models can assist with questionnaire generation and revision. Adhikari et al. combine questionnaire generation, LLM-simulated pretesting, review, and cultural adaptation within a single development pipeline [1]. Their evaluations suggest that LLM-generated or adapted wording can be useful, while additional simulated refinement does not necessarily improve an item and can sometimes increase its complexity. Fuchs et al. similarly show that the quality of LLM-generated survey items varies across models and prompting strategies [2]. Taken together, this work indicates that LLMs can support questionnaire construction, but additional model calls or more elaborate prompting cannot be assumed to improve questionnaire quality.

A second line of work focuses more directly on identifying problems in existing questions. Metheney and Yehle examine free-text LLM critiques and find that the number and type of reported problems vary with model and prompting condition [6]. Sturgis et al. evaluate LLMs against deliberately embedded questionnaire flaws and clean control items, showing that guided prompting can detect a substantial proportion of known problems while still exhibiting model- and prompt-dependent

differences [10]. Roberts et al. move further toward structured appraisal by comparing LLM-based QAS-99 decisions with expert and novice human coders; agreement varies substantially by appraisal category, and the models identify fewer problems than the questionnaire-design expert [8]. Across these studies, LLM performance is therefore not uniform across models, prompts, or types of questionnaire problem.

The existing literature establishes that LLMs can generate questionnaire items, provide useful critiques, simulate selected pretesting activities, and apply structured appraisal criteria. What is less well established is whether the same system can reliably combine several requirements: recognize when no intervention is needed, recover all independently supported defects when several occur together, avoid unsupported diagnoses, and convert the detected problems into minimal construct-preserving revisions. The present study evaluates these requirements jointly under one controlled benchmark while also comparing prompt guidance and runtime organization. A more detailed comparison with selected prior systems and studies is provided in Appendix B.

3 Benchmark and Task Design

3.1 Questionnaire-Defect Taxonomy

The benchmark uses a closed 16-label taxonomy of questionnaire-design defects developed for the present study. The taxonomy operationalizes questionnaire-design problems and principles discussed by Saris and Gallhofer [9], while the specific labels and decision boundaries were defined for consistent annotation and evaluation in this study. Let $\mathcal{T}$ denote this taxonomy and $Y_i \subseteq \mathcal{T}$ the gold label set for item i . A clean item has $Y_i = \emptyset$; a flawed item has one or more labels. Multiple labels are assigned only when each defect has separate visible evidence, fixing either defect alone would leave the other unresolved, and each would independently justify a repair. This independent-label rule prevents stylistic preferences or correlated symptoms from becoming additional gold labels.

Table 1 maps the labels to the five repair families used by the orchestrated system. Counts are label appearances, not unique items; a multi-label item contributes once for every applicable label.

Table 1: Taxonomy labels, repair families, and gold-label appearances.

Repair family	Taxonomy labels	Instances
Wording clarity	leading_question, loaded_question, recall_error, vague_ambiguous, negative_wording	65
Construct alignment	double_barreled	12
Bias sensitivity	sensitive_topic_direct, social_desirability	24
Questionnaire format	open_closed_mismatch	12
Response options and scale	agree_disagree_scale, unbalanced_scale, incomplete_options, non_exclusive_options, missing_scale_labels, too_many_scale_points, polarity_mismatch	93
Total	16 taxonomy labels	206

Several of these distinctions correspond to questionnaire-design problems discussed by Saris and Gallhofer, including double-barreled requests, implicit assumptions, response-category coverage and exclusivity, scale construction, and stem–response compatibility [9].

To connect defect detection with revision, the 16 taxonomy labels are organized into five specialist families according to the type of repair they require. Taxonomy labels remain the units of annotation and evaluation, whereas specialist families determine which role is responsible for revising an item.

Wording clarity. This family covers defects that affect how respondents interpret the question stem. `leading_question` identifies wording that steers respondents toward a particular answer, whereas `loaded_question` identifies an unverified presupposition. `recall_error` concerns demands that exceed reasonable memory capabilities, while `vague_ambiguous` captures unclear or underspecified meaning. `negative_wording` covers difficult negation or reverse wording.

Construct alignment. This family contains `double_barreled`, which applies when a single response is required for two separable constructs that respondents could evaluate differently.

Bias sensitivity. This family distinguishes two sources of bias. `sensitive_topic_direct` concerns insufficient protection when asking about sensitive matters, whereas `social_desirability` concerns wording that pressures respondents toward a socially approved answer.

Questionnaire format. This family contains `open_closed_mismatch`, which identifies an incompatibility between the response task requested by the question stem and the format of the supplied response options.

Response options and scale. This family covers defects in how answers are represented. `agree_disagree_scale` identifies the use of generic agreement options when an item-specific scale is required. `incomplete_options` concerns inadequate response coverage, `non_exclusive_options` concerns overlapping categories, and `unbalanced_scale` concerns asymmetric coverage of an ordered continuum. The remaining labels address scale construction: `missing_scale_labels` concerns interpretability, `too_many_scale_points` concerns unjustified precision, and `polarity_mismatch` concerns disagreement between the question stem and the direction of the response scale.

The five families contain five, one, two, one, and seven labels, respectively. This unequal structure reflects similarities in the required repair operations rather than an attempt to balance the groups. Their benchmark frequencies also vary substantially, ranging from 12 to 93 label appearances. Family-level scores should therefore be interpreted with their denominators and alongside the corresponding per-label results.

3.2 Final 200-Item Gold Benchmark

The empirical evaluation uses an expert-reviewed gold benchmark of 200 questionnaire items. The benchmark was constructed through a human-directed, LLM-assisted process in which frontier large language models supported item drafting, diversification, and construct-preserving revision. The taxonomy, benchmark design, annotation criteria, and final labels were defined and reviewed by the researchers. The resulting benchmark therefore serves as an expert-reviewed reference standard for evaluating defect detection and questionnaire revision.

Table 2: Composition of the final expert-reviewed benchmark.

Property	Count
Total questionnaire items	200
Clean controls	40
Flawed items	160
Single-label flawed items	115
Multi-label flawed items	45
Taxonomy labels	16
Total gold-label appearances	206

The 40 clean controls contain no independently supported defect under the taxonomy. Their inclusion is essential because a questionnaire-review system must do more than detect problematic items: it must also recognize when no intervention is required. These items therefore make it possible to measure false-positive detection and unnecessary revision, or *overcorrection*. For clean controls, the

reference revision preserves the original question and response options unchanged.

The 160 flawed items are divided into 115 single-label and 45 multi-label cases. A single-label item isolates one principal defect mechanism and is useful for determining whether a model can distinguish closely related taxonomy categories. A multi-label item contains two or more defects that are independently supported by visible evidence. In such cases, correcting only one defect is insufficient because another measurement problem remains. These items therefore test whether a system can recover a complete label set and coordinate a revision that addresses all applicable problems while preserving unaffected parts of the item. Because multi-label items contribute once to the count of each applicable category, the 200 items contain 206 gold-label appearances.

The benchmark spans 14 substantive domains, nine response formats, and three approximately balanced difficulty groups; detailed topic counts are reported in Appendix B.

Taken together, the benchmark supports three complementary forms of evaluation. Clean controls test restraint, single-label items test category-specific discrimination, and multi-label items test completeness and coordinated repair. Its breadth of topics, response formats, and difficulty levels further tests whether questionnaire-defect detection transfers across different substantive contexts rather than depending on one domain or a small set of surface patterns. At the same time, the benchmark remains a controlled, LLM-assisted and expert-reviewed collection rather than a representative sample of all questionnaires used in practice; this limitation is considered in Section 5.7.

3.3 Detection and Revision Tasks

For item i , the benchmark input is $x_i = (q_i, O_i)$, with question q_i and ordered options O_i . Only these fields enter the initial checker or router. Downstream roles may receive runtime-generated issues or plans, but never IDs, gold labels, expected revisions, target concepts, topics, metadata, or review notes; direct annotation and revision leakage is therefore excluded.

The detection task is multi-label classification,

$$\hat{Y}_i = f(q_i, O_i), \quad \hat{Y}_i \subseteq \mathcal{T}.$$

The model must return the empty set for a clean item and every independently supported label for a flawed item; omissions and unsupported additions are both errors. The revision task produces $(\hat{q}_i, \hat{O}_i)$ and must fix all detected defects with minimal change while preserving the intended construct. A multi-label repair is incomplete if one defect remains, whereas a clean item must be returned exactly unchanged. Each flawed row has one reference revision for reproducible comparison; semantic editing metrics supplement the strict reference-based measures.

4 Methodology

The study varies two components independently: the runtime architecture and the prompt treatment. The baseline and orchestrated architectures are each evaluated with the additive P0–P2 prompt conditions. Within an experimental run, the same LLM checkpoint is used for every role; orchestration changes the division of responsibilities and sequence of calls rather than combining different models.

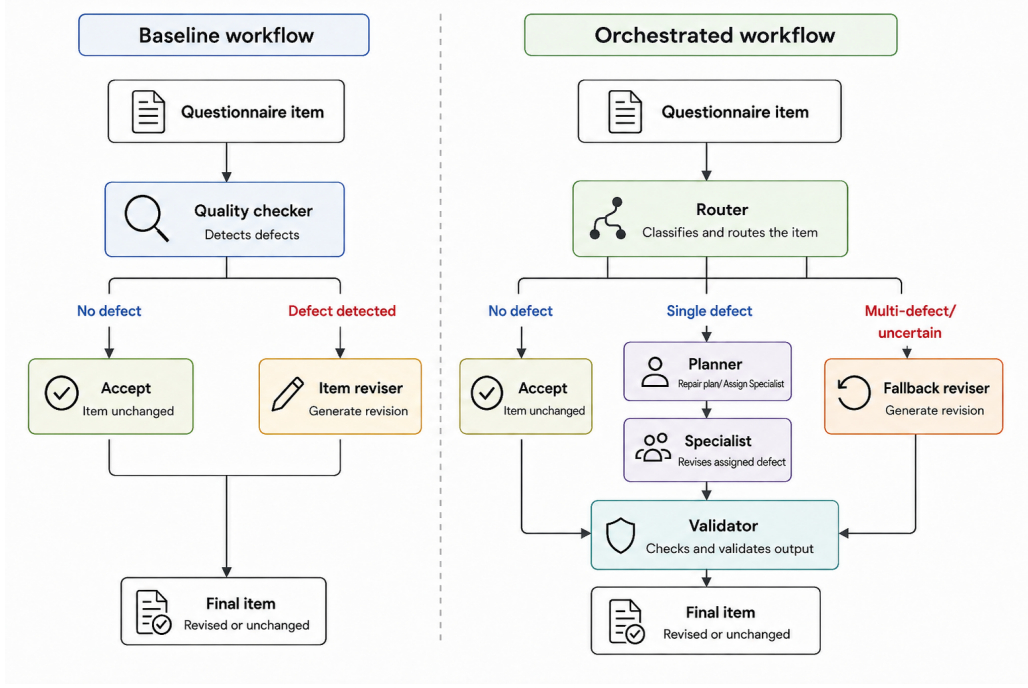


Figure 1: High-level comparison of the baseline and orchestrated workflows. Detailed role-level control flow is provided in Appendix A, Figures 2 and 3.

4.1 Baseline Architecture

The baseline architecture consists of an LLM-based quality checker followed, when necessary, by an LLM-based item reviser. For an item $x = (q, O)$, where q is the question and O the ordered response-option list, the checker predicts all independently supported taxonomy defects and returns their categories, item-specific severity, explanations, and evidence. It does not revise the item.

The next step depends on the checker output:

$$\text{Item} \rightarrow \text{Quality Checker} \rightarrow \begin{cases} \text{preserve unchanged,} & \text{no detected defect,} \\ \text{general Item Reviser,} & \text{one or more detected defects.} \end{cases}$$

The reviser receives the original item and the detected issues and returns a revised question, response options, revision notes, and a changed flag. When no issue is detected, revision is skipped and the original item is copied to the final output. The baseline therefore provides a simple reference architecture without explicit routing, specialist assignment, or post-revision validation. The baseline does not use the repair-family decomposition. Its two general roles—the quality checker and item reviser—are both prompted against the full 16-label taxonomy, whereas repair-family-specific roles are used only in the orchestrated architecture.

4.2 Orchestrated Architecture

The orchestrated architecture separates detection, planning, repair, and validation into role-specific LLM calls. Its control flow branches after the router:

$$\text{Item} \rightarrow \text{Router} \rightarrow \begin{cases} \text{unchanged item} \rightarrow \text{Validator,} & \text{clean accept path,} \\ \text{Planner} \rightarrow \text{Specialist} \rightarrow \text{Validator,} & \text{clear single-label path,} \\ \text{Fallback Reviser} \rightarrow \text{Validator,} & \text{multi-label or uncertain path.} \end{cases}$$

Routing and planning. The router returns `accept`, `revise`, or `fallback`, together with taxonomy labels, evidence, a confidence score, and a recommended route. A single clearly supported defect is mapped to its canonical specialist family. The default policy instead uses fallback for multiple labels, mixed families, confidence below 0.7, contradictory outputs, unsupported labels, or ambiguous routes. The planner is invoked only on the specialist path. It converts the router’s fixed issue set into a minimal repair plan but may neither change the labels nor rewrite the item.

Specialists and fallback. The five specialist roles correspond to the repair families defined above: wording clarity, response options and scale, construct alignment, bias sensitivity, and questionnaire format. Each specialist repairs only its assigned family and must preserve all unrelated content. The fallback reviser handles multi-label, low-confidence, conflicting, or otherwise unsafe cases. It coordinates all independently supported repairs when the evidence is clear, but may preserve the original item when a speculative change would be less safe.

Validation. The validator evaluates rather than rewrites. It checks whether the candidate preserves the construct, fixes every supplied defect, and introduces no new supported defect. Clean items accepted by the router are also validated in unchanged form. The possible statuses are `pass`, `retry`, `manual_review`, and `failed`. One validator-triggered retry is permitted; unresolved or unsafe candidates are flagged for manual review. `failed` is reserved for outputs that cannot be evaluated because the candidate information is missing, malformed, or unusable.

All roles return schema-constrained JSON. Invalid JSON or schema violations trigger format-repair attempts within the same role; these are separate from the validator’s semantic revision retry.

4.3 Prompt Treatments and Revision Contract

The prompt conditions are additive:

$$P0 \subset P1 \subset P2.$$

P1 retains P0 and adds operational procedures, while P2 retains P1 and adds fixed demonstrations. Separate but corresponding prompt packs are used for the baseline and orchestrated architectures.

P0 is the zero-shot control. It provides definitions of all 16 labels, decision boundaries between related categories, an evidence gate, independent multi-label logic, clean-item preservation, and minimal-revision rules. It contains no demonstrations and therefore measures performance from the codebook and task instructions alone.

P1 was introduced after preliminary development runs showed weaknesses on response-option, scale, and questionnaire-format defects. In addition to P0, it requires the model to identify the response mode and intended dimension or unit before checking agreement proxies, completeness, mutual exclusivity, balance, labels, granularity, polarity, count-rate consistency, and open-closed compatibility. Matching repair procedures require the smallest supported correction and prohibit routine additions such as “Other”, “Don’t know”, or refusal options without visible evidence. P1 contains no few-shot examples, so it isolates the effect of explicit operational guidance. In the orchestrated pipeline, these additions are concentrated in the router, fallback reviser, response-options specialist, questionnaire-format specialist, and validator.

P2 retains P1 and adds fixed, role-specific, schema-valid demonstrations. The examples calibrate detection, revision, routing, fallback restraint, clean-item preservation, and validation. Every specialist family is represented, but the examples are not balanced across all 16 individual taxonomy labels:

$$P2 \text{ is family-complete, but not taxonomy-complete.}$$

P2 contains 37 fixed, role-specific demonstrations spanning all five specialist families; the full role-wise distribution is reported in Appendix C.

All prompt conditions treat revision as targeted repair rather than unrestricted rewriting. The model must correct only independently supported defects while preserving the construct, population, reference period, response dimension, and all non-defective wording and option properties. The question or response set should be changed only where required, clean items must remain unchanged, and no repair may introduce a new taxonomy-defined problem. For multi-label items, every independently supported label remains a separate repair obligation.

4.4 Thinking-Mode Ablation

A supplementary P2 ablation evaluates the Qwen model’s native reasoning, or “thinking”, mode under both architectures. The prompts and external JSON contract remain unchanged; only the model’s chat-template thinking mechanism is toggled. When enabled, the internal reasoning segment is removed before the final response is parsed against the same schema as in the non-thinking condition. Completion reliability and conditional task performance are reported separately in Section 5.6.

5 Results and Discussion

The experiments are analyzed descriptively because each condition was run once with deterministic decoding. The main comparisons therefore concern matched changes in models, prompt packs, and architectures rather than sampling-based significance tests. Detection results are reported on the complete benchmark, whereas revision metrics are computed on the 160 gold-flawed items with valid reference revisions. Failed items count as missing predictions in detection metrics and are excluded from semantic revision averages.

5.1 Analysis Scope and Completion Reliability

All 20 non-thinking runs reached all 200 benchmark items. Most also showed high structured-output reliability: across the twelve non-thinking Qwen runs, only one item-level failure occurred. Mistral produced ten failures across its four P0 runs, and Gemma produced 17 across its four P0 runs.

The four thinking-mode runs are treated separately in Section 5.6. Three stopped before all 200 items had been reached, and all four produced substantial numbers of invalid structured outputs, mostly because internal reasoning exhausted the available generation budget before a valid JSON answer was completed. Consequently, their task metrics are reported only conditionally on successful outputs. Whenever a thinking result is contrasted with a non-thinking result, the non-thinking metric is recomputed on exactly the same successful item IDs.

5.2 Overall Detection Performance and Cross-Model Patterns

Table 3 reports the complete non-thinking end-to-end results. The strongest complete configuration was Qwen P2 with orchestration, which achieved micro-F1 0.563, exact label-set match 0.430, and a clean false-positive rate of 0.050. Qwen P2 with the baseline architecture achieved the highest recall, 0.684, but its lower precision and higher clean false-positive rate illustrate the baseline’s more aggressive labeling behavior.

Table 3: End-to-end detection results without thinking. Failed items remain in the 200-item denominator and cannot count as exact matches.

Model	Prompt	Architecture	Precision	Recall	F1	Exact	Clean FPR	Overcorr.	Failed
Mistral-7B	P0	Baseline	0.178	0.146	0.160	0.125	0.750	0.525	3
Mistral-7B	P0	Orchestrated	0.235	0.019	0.036	0.205	0.000	0.000	1
Gemma-2-9B	P0	Baseline	0.379	0.320	0.347	0.270	0.725	0.725	0
Gemma-2-9B	P0	Orchestrated	0.383	0.150	0.216	0.305	0.075	0.125	7
Qwen3.5-9B	P0	Baseline	0.428	0.660	0.519	0.290	0.150	0.150	0
Qwen3.5-9B	P0	Orchestrated	0.795	0.340	0.476	0.370	0.050	0.050	0
Qwen3.5-9B	P1	Baseline	0.413	0.636	0.501	0.280	0.075	0.075	1
Qwen3.5-9B	P1	Orchestrated	0.845	0.345	0.490	0.375	0.025	0.000	0
Qwen3.5-9B	P2	Baseline	0.443	0.684	0.538	0.270	0.175	0.150	0
Qwen3.5-9B	P2	Orchestrated	0.845	0.422	0.563	0.430	0.050	0.050	0

At P0, Qwen achieved the strongest performance in both architectures, followed by Gemma and then Mistral. For the weaker models, orchestration primarily increased conservatism rather than improving label recognition. Mistral, for example, produced no false positives on the 40 clean items but recovered only four of the 206 gold-label appearances. A low clean false-positive rate can therefore be misleading when it results from broad under-detection. The complete family-level breakdown is reported in Appendix D.

Conservative routing did not, however, always indicate a failure to detect defects. In some cases, it correctly prevented unnecessary revision of clean items. This distinction is illustrated by the following Qwen P2 example:

Question: *Some people prefer not to answer questions about gambling. During the last 12 months, how many times did you gamble with money?*

Response options:

0 times
1 time
2–3 times
4–11 times
12 or more times
Prefer not to answer

Gold annotation: Clean item; no taxonomy defect.

The baseline incorrectly predicted `incomplete_options` and `loaded_question` and consequently rewrote the item as a two-part filter question. By contrast, the orchestrated router detected no defect and preserved the original item. In this case, orchestration’s lower false-positive tendency reflected appropriate restraint rather than under-detection. Taken together, these results show that clean-item performance must be interpreted alongside defect recall: conservative behavior may either preserve a valid item or conceal a failure to identify genuine defects.

5.3 Effects of the P1 and P2 Prompt Packs

The matched Qwen comparisons in Table 3 isolate prompt-pack effects within each architecture. They show that additional prompt content was not automatically beneficial. P1 reduced false positives on clean items, but did not improve overall baseline detection and produced only a small orchestrated gain. P2 had a clearer effect, particularly under orchestration. Complete TP, FP, and FN diagnostics are reported in Appendix D.

P1: Operational rules. As shown in Table 3, for the baseline architecture, P1 reduced F1 from 0.519 to 0.501: true positives fell by five, false positives increased by four, and false negatives increased by five. Its main benefit was restraint on clean controls, where FPR fell from 0.150 to 0.075. For orchestration, P1 increased precision from 0.795 to 0.845 while recall remained almost unchanged, producing a modest F1 increase from 0.476 to 0.490 and reducing clean FPR to 0.025.

The operational rules helped selected categories but did not solve the broader response-option problem. As shown in the per-label results in Table 13, baseline `open_closed_mismatch` F1 improved from 0.556 to 0.632, and `unbalanced_scale` improved from 0.250 to 0.435. Conversely, `non_exclusive_options` fell from 0.545 to 0.200 and `polarity_mismatch` from 0.786 to 0.667. At the family level, Table 4 shows that response-options-and-scale F1 fell from 0.488 to 0.444 in the baseline and was essentially unchanged under orchestration (0.350 to 0.344). P1 therefore improved some decision boundaries and clean-item restraint, but explicit procedures alone did not yield a general response-option gain.

P2: Few-shot calibration. P2 produced the best Qwen F1 in both architectures. Relative to P1, baseline F1 increased from 0.501 to 0.538, driven primarily by recall. Under orchestration, P2 increased true positives from 71 to 87 while adding only three false positives; recall rose from 0.345 to 0.422, F1 from 0.490 to 0.563, and exact match from 0.375 to 0.430. Relative to P0, the orchestrated F1 gain was 0.087. The few-shot treatment was therefore most effective when demonstrations were combined with role-specific routing and repair responsibilities.

The family-level results in Table 4 show that this gain was not uniform. Under orchestration, the largest P2 improvement was in response options and scale, whose F1 increased from 0.350 under P0 and 0.344 under P1 to 0.507. Wording clarity also increased to 0.574. Construct alignment returned to 0.857, while questionnaire format remained at 0.400 and bias sensitivity declined to 0.634. In the baseline, P2 reached the highest construct-alignment F1 (0.957) and response-options F1 (0.504), but its questionnaire-format result fell below both P0 and P1. Family-spanning examples therefore supported broader transfer, but did not make performance uniformly better across families.

Table 4: Qwen family-level micro-F1 across prompt packs.

Prompt	Architecture	Wording	Construct	Bias	Format	Options/scale
P0	Baseline	0.469	0.917	0.682	0.556	0.488
P0	Orchestrated	0.495	0.857	0.667	0.286	0.350
P1	Baseline	0.474	0.870	0.698	0.632	0.444
P1	Orchestrated	0.547	0.800	0.684	0.400	0.344
P2	Baseline	0.503	0.957	0.698	0.471	0.504
P2	Orchestrated	0.574	0.857	0.634	0.400	0.507

Complete per-label F1 values are reported in Appendix D (Table 13). P2 orchestration produced especially strong gains for `leading_question` and `loaded_question`, and it improved several response-option and scale defects, including agreement proxies, incomplete option sets, missing scale labels, excessive scale points, and polarity mismatches.

The improved distinction between leading and loaded wording is visible in one representative item. The benchmark question was:

Question: *Since local arts venues bring communities together, should public funds be used to support them?*

Response options:

Yes

No

Gold annotation: `leading_question`

Under P0 and P1 orchestration, the router instead predicted `loaded_question`. Under P2, it returned the correct `leading_question` label and revised the stem to “*Should public funds be used to support local arts venues?*” The example therefore illustrates the improvement in this particular decision boundary under few-shot calibration. The generated revision removed the steering cue, although it retained the original binary response format rather than the five-point support–oppose scale used in the reference revision.

These gains were nevertheless selective. Across all Qwen prompt packs, the orchestrated router produced no true positives for `recall_error`, `non_exclusive_options`, or `unbalanced_scale`. Family-spanning examples therefore improved selected labels without guaranteeing taxonomy-complete performance: P2 was family-complete, but not taxonomy-complete.

5.4 Architecture Trade-Offs and Multi-Label Detection

The architecture comparison is especially visible in the cardinality of the predicted label sets. As Table 5 shows, the baseline checker tended to produce broad predictions. Across P0–P2, it returned two or more labels for 114–128 of the 200 items, while returning exactly one label for only 5–10 items. This behaviour contributed to the baseline’s comparatively high recall and its ability to recover multiple gold defects, but it also produced many unsupported additions. The difference between *Multi all-gold* and *Multi exact* illustrates this trade-off: under P2, the baseline recovered every gold label for 24 of the 45 multi-label items, but only 16 predictions remained exact after additional predicted labels were taken into account.

The orchestrated router showed the opposite prediction pattern. In all three non-thinking Qwen conditions, every output contained either zero or one taxonomy label; no prediction contained two or more labels. This coincided with substantially stronger restraint on clean and single-label items, but also with incomplete recovery of multi-label cases. Under P2, orchestration recovered at least one correct label on 39 of the 45 multi-label items, yet recovered the complete gold label set on none. The corresponding baseline recovered at least one correct label on 41 items and all gold labels on 24.

Table 5: Prediction cardinality and exact label-set recovery for Qwen. The columns 0, 1, and ≥ 2 count predicted label-set sizes.

Prompt	Architecture	0	1	≥ 2	Clean exact	Single exact	Multi exact	Multi any	Multi all-gold
P0	Baseline	68	10	122	34/40	10/115	14/45	39/45	24/45
P0	Orchestrated	112	88	0	38/40	36/115	0/45	34/45	0/45
P1	Baseline	80	6	114	37/40	6/115	13/45	39/45	26/45
P1	Orchestrated	116	84	0	39/40	36/115	0/45	35/45	0/45
P2	Baseline	67	5	128	33/40	5/115	16/45	41/45	24/45
P2	Orchestrated	97	103	0	38/40	48/115	0/45	39/45	0/45

This distinction is visible when the 45 gold multi-label items are examined directly. Under P0 and P1, the orchestrated router returned exactly one label for 36 items and no label for nine. Under P2,

it returned one label for 41 items and no label for four. None of the three conditions produced two or more labels for a gold multi-label item. Thus, although P2 increased the number of multi-label items for which at least one correct defect was recovered, it did not produce complete defect-set recovery.

A multi-label item illustrates the downstream consequence of this detection pattern. The benchmark item asked:

Question: *To what extent do you disagree that exam instructions were not unclear?*

Response options:

Strongly disagree
Somewhat disagree
Neither agree nor disagree
Somewhat agree
Strongly agree

Gold annotation: `negative_wording`, `agree_disagree_scale`

Under P2, the baseline recovered both gold labels and replaced the agreement options with the reference scale. The orchestrated router returned only `negative_wording`; its revision simplified the stem to “*To what extent do you agree that exam instructions were unclear?*” but retained the generic agreement options. Thus, the missed secondary diagnosis corresponded to a repair obligation that never reached the downstream revision stage.

The fallback-route counts are consistent with this pattern. Across the complete 200-item benchmark, fallback was used only 3, 3, and 4 times under P0, P1, and P2, respectively. These fallback routes resulted from planner or validator control decisions rather than from successful recognition of multiple defects by the router. Consequently, the end-to-end results provide little evidence about how the fallback reviser would behave when supplied with a complete multi-label diagnosis.

Overall, the cardinality results clarify the precision–recall difference between the two architectures. The baseline frequently produced broad defect sets, which increased multi-label coverage but also introduced additional false positives. The orchestrated router was substantially more selective, improving clean-item and single-label exact-match performance while often recovering only part of the gold defect set on multi-label items. The observed multi-label limitation was therefore associated with upstream detection behaviour rather than with an inability of the downstream orchestration mechanism to represent or route multiple labels.

5.5 Revision Quality and Exact Response-Option Matching

Table 6 reports end-to-end revision metrics. The BERTScore values are tightly concentrated, especially for Qwen, because the original, generated, and reference questions usually share most of their text. SARI is more discriminating, but it also shows that improved detection did not automatically produce a higher reference-based revision score. Qwen P2 orchestration had the best detection F1, yet its SARI of 60.96 was below the P0 baseline value of 62.45.

Table 6: End-to-end revision metrics without thinking. Exact-option matching requires the complete normalized option list in the reference order. Changed is computed on successfully scored flawed items.

Model	Prompt	Architecture	BERTScore	SARI	Exact Q.	Exact options	Exact full	Changed	Scored
Mistral-7B	P0	Baseline	0.951	52.33	0.063	0.089	0.000	0.595	158/160
Mistral-7B	P0	Orchestrated	0.952	58.94	0.164	0.113	0.000	0.107	159/160
Gemma-2-9B	P0	Baseline	0.955	56.93	0.119	0.094	0.013	0.894	160/160
Gemma-2-9B	P0	Orchestrated	0.953	60.98	0.183	0.105	0.013	0.529	153/160
Qwen3.5-9B	P0	Baseline	0.956	62.45	0.175	0.113	0.013	0.762	160/160
Qwen3.5-9B	P0	Orchestrated	0.956	61.15	0.175	0.113	0.019	0.537	160/160
Qwen3.5-9B	P1	Baseline	0.956	62.44	0.182	0.082	0.006	0.717	159/160
Qwen3.5-9B	P1	Orchestrated	0.954	60.42	0.175	0.113	0.013	0.519	160/160
Qwen3.5-9B	P2	Baseline	0.956	61.89	0.188	0.150	0.019	0.781	160/160
Qwen3.5-9B	P2	Orchestrated	0.955	60.96	0.175	0.125	0.019	0.619	160/160

The exact-option metric provides the clearest revision-side P2 signal. For Qwen baseline, exact response-option matching increased from 18 of 160 items under P0 (0.113) and 13 of 159 under P1 (0.082) to 24 of 160 under P2 (0.150). For orchestration, it increased from 18 of 160 under P0 and P1 to 20 of 160 under P2. This is consistent with P2’s direct calibration of response-scale and option repairs. Nevertheless, even the best complete end-to-end condition matched the reference option list on only 15% of flawed items. Exact full revision never exceeded 1.9% in the complete end-to-end Qwen runs, confirming that a single reference is a strict target and that semantically acceptable alternatives often differ in wording, option labels, or order.

For the weaker models, orchestration improved SARI despite reducing detection recall. Mistral increased from 52.33 to 58.94 and Gemma P0 from 56.93 to 60.98. This suggests that when these models did reach the revision stage, role separation constrained edits more effectively than the general reviser. It does not imply better end-to-end defect coverage, because their orchestrated changed rates were only 0.107 and 0.529, respectively, and many flawed items were left unchanged.

Oracle revision was not a strict empirical upper bound. Correct categories were supplied without item-specific evidence, so conservative revisers sometimes rejected a repair and left gold-flawed items unchanged. Exact-full revision also remained very low, peaking at 0.025 (4 of 160 items) in the complete non-thinking oracle runs. Complete oracle metrics and diagnostics are reported in Appendix E.

Overall, the revision results support three conclusions. First, P2 improved reference-exact response options more clearly than it improved question-level semantic scores. Second, detection improvements and revision improvements were not interchangeable: the strongest detector did not maximize SARI. Third, BERTScore, SARI, and exact matching capture complementary properties but none establishes that every defect was repaired, the construct was preserved, and no new defect was introduced.

5.6 Thinking-Mode Analysis

Thinking mode had poor completion reliability under the available token budget: internal reasoning frequently exhausted the generation allowance before a valid JSON response was completed. Detailed completion counts are reported in Appendix F.

The conditional end-to-end detection comparison in Table 7 uses only the exact IDs for which the thinking run returned a valid structured output. On these matched subsets, thinking improved both architectures. Baseline F1 increased from 0.460 to 0.727, exact match from 0.351 to 0.789, and clean FPR fell from 0.143 to 0.048. Orchestrated F1 increased from 0.686 to 0.784, mainly through a recall increase from 0.545 to 0.659, while clean FPR remained zero.

Table 7: Thinking versus non-thinking P2 detection on identical thinking-successful item IDs.

Architecture	Condition	n	Precision	Recall	F1	Exact	Clean FPR	Overcorr.
Baseline	Thinking	57	0.737	0.718	0.727	0.789	0.048	0.048
Baseline	Non-thinking	57	0.377	0.590	0.460	0.351	0.143	0.143
Orchestrated	Thinking	62	0.967	0.659	0.784	0.758	0.000	0.000
Orchestrated	Non-thinking	62	0.923	0.545	0.686	0.677	0.000	0.000

These matched results indicate that reasoning improved performance conditional on successful outputs, but they do not establish a full-benchmark advantage. The successful baseline subset contained only three multi-label items and the orchestrated subset only one, compared with 45 in the full benchmark. Successful outputs were therefore heavily biased toward clean and single-label cases.

The only thinking-mode semantic revision result is reported in Appendix F. Because no matched non-thinking semantic rescore was archived, it does not support a direct revision comparison. Thinking mode was thus promising conditionally but operationally unreliable.

5.7 Limitations and Implications

The findings should be interpreted as evidence about the behaviour of the evaluated systems under a controlled questionnaire-appraisal task, rather than as a direct estimate of performance on questionnaires encountered in practice. The benchmark was deliberately constructed to make clean items, isolated defects, and independently supported multi-label cases observable under a fixed 16-label taxonomy. This provides experimental control and makes specific failure modes measurable, but it also narrows the domain in which the reported scores have meaning. Real questionnaires may contain defects that fall outside the taxonomy, depend on survey context or target-population knowledge that is not visible in a single item, or involve ambiguous cases for which experts themselves would disagree. In addition, although the benchmark spans multiple topics and response formats, it is an expert-reviewed, LLM-assisted collection of 200 items rather than a representative sample of deployed survey instruments. The results should therefore be interpreted as controlled comparative evidence rather than population-level estimates of questionnaire-appraisal accuracy.

Revision quality is particularly difficult to reduce to a single automatic score. Each flawed item has one reference revision, although several different revisions may be methodologically acceptable. Exact question, option, and full-item matching consequently reward agreement with one particular realization rather than revision quality in the broader sense. Semantic metrics alleviate this problem only partially: a high BERTScore may reflect lexical similarity while leaving the original defect unresolved, whereas SARI rewards particular patterns of addition, deletion, and retention without directly establishing construct preservation. The revision results are therefore most informative when the metrics are considered jointly. None of the reported automatic measures alone demonstrates that every defect was repaired, that no new defect was introduced, and that the intended construct was fully preserved.

Taken together, these limitations suggest a more qualified interpretation of the architecture comparison. The baseline and orchestrated systems exhibit different error profiles rather than a simple ordering in which one dominates the other. The baseline more readily proposes multiple defects and consequently achieves higher recall, but often at the cost of unsupported labels and unnecessary revision. Orchestration improves precision, restraint, and interpretability of the repair process, but its effectiveness depends on successful routing and reliable structured interaction between roles. A stronger practical system would therefore combine these complementary properties: broad multi-label detection followed by evidence-based verification, deterministic checks where questionnaire defects can be mechanically established, role-separated revision where specialization is useful, robust schema

handling, and human review where construct preservation cannot be established from the visible item alone.

6 Conclusion and Future Work

6.1 Conclusion

This study examined whether locally deployed LLMs can identify taxonomy-defined questionnaire defects and convert those diagnoses into targeted revisions while preserving acceptable items. The evaluation combined a 200-item expert-reviewed benchmark with two runtime architectures, three additive prompt conditions, multiple model families, end-to-end and oracle-revision modes, and a supplementary thinking-mode experiment. Rather than identifying a single universally best configuration, the results reveal distinct trade-offs between defect coverage, restraint, revision behaviour, and operational reliability.

The clearest architectural difference was the error profile produced by the two pipelines. The baseline checker behaved as a comparatively broad detector: it recovered more gold defects and more often proposed multiple labels, but also generated substantially more unsupported predictions. The orchestrated system was considerably more selective. Its routing and validation stages improved precision and clean-item preservation, but this restraint also reduced recall. The strongest complete configuration was Qwen using the orchestrated workflow with family-spanning few-shot calibration, which achieved a detection micro-F1 of 0.563, an exact label-set match rate of 0.430, and a clean false-positive rate of 0.050. These results therefore do not imply that orchestration simply dominates the baseline. Instead, the architectures place errors in different locations: the baseline risks over-diagnosis, whereas orchestration risks failing to surface defects before they can reach the appropriate repair stage.

The prompt-pack experiments similarly show that more guidance is not automatically better. Adding explicit operational response-option and questionnaire-format rules mainly improved restraint and selected response-format decisions rather than producing a general detection gain. Adding family-spanning few-shot demonstrations yielded the strongest Qwen results and produced clear improvements for several wording, response-option, and scale defects, particularly under orchestration.

Thinking mode showed the same difference between conditional capability and end-to-end usefulness. On the subsets for which valid structured outputs were completed, reasoning substantially improved detection performance. Yet frequent token exhaustion prevented many items from reaching a valid final JSON output. The experiment therefore provides evidence that additional reasoning can help the underlying decision task, but not that the evaluated thinking configuration was operationally superior.

Taken together, the study supports a more qualified view of LLM-assisted questionnaire appraisal. LLMs can provide useful screening and revision support, but their behaviour depends strongly on how the task is decomposed, how defects are operationalized, and how model outputs are constrained. Architectural control can improve restraint and make failures more interpretable, while prompt calibration can improve selected decisions, but neither substitutes for complete defect detection or methodological judgment. The most promising role for such systems is therefore as structured assistance for questionnaire review rather than as an autonomous replacement for expert appraisal and respondent-based pretesting.

6.2 Future Work

The most immediate technical priority is to improve multi-label detection before revision is attempted. Because the current implementation can already pass multiple detected labels to the fallback reviser,

the central problem observed in the experiments lies upstream: the router often identified one supported defect while overlooking another. A natural extension would be an explicit second-defect verification step after the first diagnosis, or a separate multi-label detection stage that asks whether any independently supported defect remains unexplained. Multi-label outputs could also be represented with label-specific evidence and rationales rather than one shared evidence field, giving the fallback reviser a clearer repair obligation for each detected problem.

The experimental comparison should also be broadened. Complete P0–P1–P2 comparisons across additional model families would help determine which effects are specific to Qwen and which generalize across checkpoints. Repeated runs under alternative seeds or decoding settings would further distinguish stable architecture and prompt effects from run-specific behaviour. External evaluation on questionnaire items drawn from deployed instruments would also be valuable, particularly for defects that depend on survey context, target population, or information not visible in an isolated item.

Finally, revision evaluation requires stronger evidence than a single reference formulation can provide. Exact matching remains useful because it is transparent and easy to reproduce, but it penalizes semantically acceptable alternatives. Future work could therefore combine multiple expert reference revisions, targeted human evaluation of construct preservation and defect resolution, and validated semantic comparison of response options. Appendix G outlines an exploratory SOM-F1 formulation for the latter purpose, but such a metric should only be incorporated into the main evaluation after independent human-labelled validation.

6.3 AI use statement

LLMs were used only for language and grammatical editing (at the sentence level). All research design, analysis, and interpretation were conducted exclusively by the author.

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A Supplementary Architecture Figures

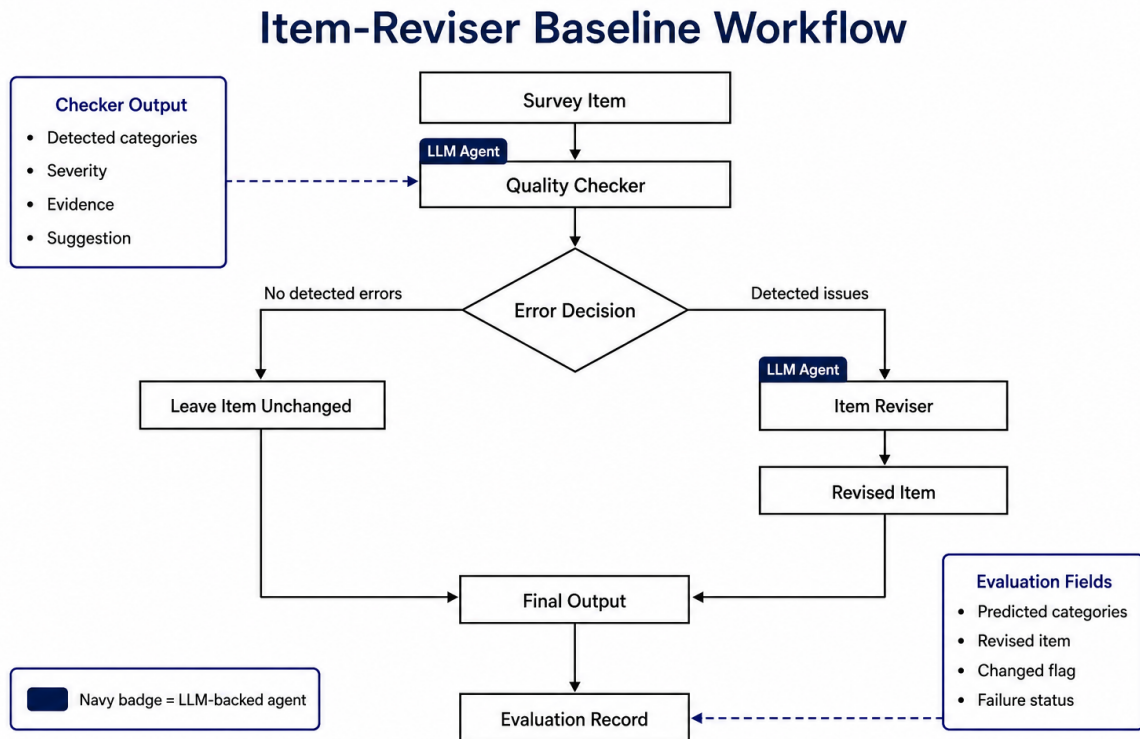


Figure 2: Baseline workflow with quality checker, item reviser.

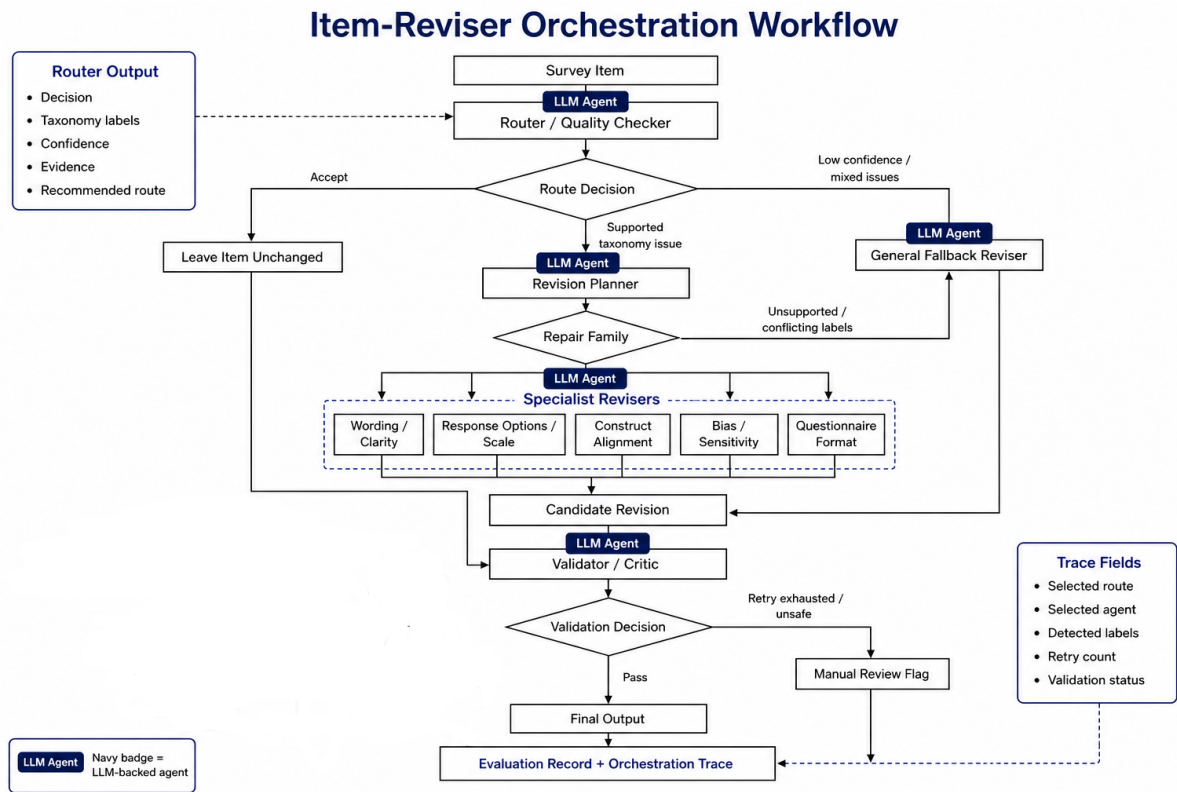


Figure 3: Orchestrated workflow with specialist routing, fallback handling, validation, and bounded retry.

B Supplementary Benchmark Information

B.1 Benchmark Topic, Format, and Difficulty Coverage

The benchmark was also designed to reduce dependence on a narrow set of questionnaire topics. It spans 14 substantive domains, as shown in Table 8. No single topic accounts for more than 19 of the 200 items.

Table 8: Topic coverage of the 200-item benchmark.

Topic	Count	Topic	Count
Education	19	Health	19
Public services	19	University life	16
Family/household	16	Technology	15
Environment	15	Politics/public policy	13
Mobility	13	Work	13
Media/culture	13	Finances	12
Labor	10	Sensitive behaviors	7

Diversity extends beyond subject matter. The items use nine response formats, including binary yes–no questions, open-ended questions, frequency scales, numeric ranges, ordinal categories, categorical response sets, support–oppose scales, agreement scales, and filter questions. The benchmark also contains three nearly equal difficulty groups: 67 **obvious** items, 67 **realistic** items, and 66 **borderline** items. The obvious cases contain comparatively salient defects, realistic cases resemble defects that may occur in ordinary questionnaire development, and borderline cases require finer distinctions between related taxonomy categories or between an actual defect and a merely optional improvement.

B.2 Detailed Comparison with Prior Work

Table 9 summarizes the closest lines of work. The comparison is selective rather than exhaustive and emphasizes differences in task definition and evaluation design.

Table 9: Selected questionnaire-evaluation and LLM studies in relation to the present work.

Study/system	Primary task	Evaluation basis	Main output	Relation to the present study
Graesser et al. [3]	Comprehension screening	Expert comparisons, assisted revision, and respondent-processing evidence	Potential comprehension problems	Automated and actionable, but concentrated on wording, syntax, semantics, and memory burden.
Saris and Gallhofer [9] / SQP	Measurement-quality prediction	MTMM-derived quality estimates and coded formal or linguistic item features	Predicted reliability, validity, and quality	Predicts expected quality for supported item types rather than explicit defect sets or generated repairs.
Adhikari et al. [1]	Generation, simulated pretesting, and cultural adaptation	Participant and expert evaluations	Generated, reviewed, and adapted questionnaire items	Evaluates a development pipeline rather than gold-taxonomy defect classification.
Metheney and Yehle [6]	Prompted critique of existing questions	262 questions in a 2×3 model-persona experiment; qualitative coding	Free-text problem statements	Studies feedback content and prompt effects without a closed gold multi-label benchmark.
Sturgis et al. [10]	Detection of deliberately embedded flaws	20 flawed questions and seven control questions	Detected problem descriptions	Includes controls and known flaws, but uses a small benchmark and does not systematically evaluate minimal revision.
Roberts et al. [8]	Automated QAS-99 appraisal	118 draft questions compared with expert and novice coders	QAS-99 binary appraisal decisions	Directly evaluates structured appraisal, but not end-to-end generated repair or exact multi-label taxonomy recovery.
Fuchs et al. [2]	LLM survey-item generation	Five models, three prompting strategies, and SQP-based evaluation	Newly generated survey items	Provides survey-specific evidence that model and prompt choices affect item quality.
Present study	Defect detection and minimal revision	200 expert-reviewed items: 40 clean, 115 single-label, and 45 multi-label	Taxonomy label set and revised question/options	Compares models, additive prompt packs, and baseline versus orchestrated architectures under one evaluation framework.

C Supplementary Prompt-Pack Information

P2 contains 37 fixed, role-specific demonstrations. The role-wise distribution is reported in Table 10.

Table 10: Distribution of demonstrations in P2.

Runtime role	Examples
Baseline quality checker / item reviser	4 / 4
Orchestration router	8
Fallback reviser	3
Wording-clarity specialist	2
Response-options-and-scale specialist	5
Construct-alignment specialist	2
Bias-sensitivity specialist	2
Questionnaire-format specialist	2
Validator	5
Revision planner	0
Total	37

The router includes at least one example for each specialist family, while the planner remains zero-shot because it only translates a fixed issue set into the canonical family and agent. All examples were authored outside the evaluation benchmark.

D Supplementary Detection Results

D.1 P0 Cross-Model Family Breakdown

At P0, Qwen clearly outperformed the other checkpoints. Its baseline F1 of 0.519 exceeded Gemma’s 0.347 and Mistral’s 0.160. Under orchestration, Qwen still reached 0.476, compared with 0.216 for Gemma and 0.036 for Mistral. The very low clean false-positive rates of the weaker orchestrated models should not be read as uniformly good performance. Mistral orchestration, for example, accepted all 40 clean controls but detected only four of the 206 gold-label appearances. Its exact-match rate of 0.205 was therefore driven almost entirely by clean items rather than successful defect classification. Similarly, Gemma P0 orchestration achieved 37 of its 61 exact matches on clean controls and recovered no multi-label item exactly.

The P0 family results in Table 11 show that the model differences were not confined to one defect type. Qwen was strongest or near strongest across all five families. Gemma’s baseline result was competitive for questionnaire format ($F_1 = 0.870$) and construct alignment ($F_1 = 0.762$), but much weaker for response options and scale ($F_1 = 0.200$). Mistral showed no true positives for construct alignment or bias sensitivity and only limited performance in the other families. Orchestration therefore did not compensate for weak underlying label recognition; it mostly made the weaker models more conservative.

Table 11: P0 family-level micro-F1 by model and architecture.

Model	Architecture	Wording	Construct	Bias	Format	Options/scale
Mistral-7B	Baseline	0.140	0.000	0.000	0.154	0.208
Mistral-7B	Orchestrated	0.100	0.000	0.000	0.000	0.000
Gemma-2-9B	Baseline	0.343	0.762	0.444	0.870	0.200
Gemma-2-9B	Orchestrated	0.276	0.609	0.207	0.154	0.061
Qwen3.5-9B	Baseline	0.469	0.917	0.682	0.556	0.488
Qwen3.5-9B	Orchestrated	0.495	0.857	0.667	0.286	0.350

D.2 Matched Qwen Detection Counts

Table 12 reports the complete matched TP, FP, and FN diagnostics across Qwen prompt packs.

Table 12: Matched Qwen detection counts and metrics across prompt packs.

Prompt	Architecture	TP	FP	FN	Precision	Recall	F1	Exact	Clean	FPR
P0	Baseline	136	182	70	0.428	0.660	0.519	0.290		0.150
P0	Orchestrated	70	18	136	0.795	0.340	0.476	0.370		0.050
P1	Baseline	131	186	75	0.413	0.636	0.501	0.280		0.075
P1	Orchestrated	71	13	135	0.845	0.345	0.490	0.375		0.025
P2	Baseline	141	177	65	0.443	0.684	0.538	0.270		0.175
P2	Orchestrated	87	16	119	0.845	0.422	0.563	0.430		0.050

D.3 Per-Label Qwen Results

Table 13 makes the concentration of the P2 effect more explicit. P2 orchestration substantially improved `leading_question`, `loaded_question`, `agree_disagree_scale`, `incomplete_options`, `missing_scale_labels`, and `polarity_mismatch`; it also achieved perfect

F1 for `too_many_scale_points`. For example, the router changed an arts-funding item from the incorrect `loaded_question` label under P0 and P1 to the correct `leading_question` label under P2. It also began detecting an agreement-proxy item about drinking-water concern and an unlabeled confidence scale that P0 and P1 had accepted.

However, several blind spots remained. The orchestrated router produced no true positives for `recall_error`, `non_exclusive_options`, or `unbalanced_scale` under any Qwen prompt pack. P2 also reduced `sensitive_topic_direct` and `negative_wording` performance relative to P1 orchestration. This result illustrates the distinction between family-complete and taxonomy-complete calibration: examples in every family did not guarantee reliable discrimination for every label within that family.

Table 13: Per-label Qwen F1. B and O denote baseline and orchestrated architectures.

Label	Gold	P0-B	P0-O	P1-B	P1-O	P2-B	P2-O
<code>leading_question</code>	12	0.595	0.600	0.537	0.857	0.585	0.957
<code>loaded_question</code>	12	0.300	0.609	0.333	0.667	0.358	0.737
<code>double_barreled</code>	12	0.917	0.857	0.870	0.800	0.957	0.857
<code>recall_error</code>	12	0.375	0.000	0.286	0.000	0.400	0.000
<code>vague_ambiguous</code>	16	0.438	0.211	0.500	0.211	0.438	0.211
<code>sensitive_topic_direct</code>	12	0.632	0.286	0.737	0.286	0.588	0.154
<code>social_desirability</code>	12	0.720	0.857	0.667	0.917	0.769	0.857
<code>negative_wording</code>	13	0.889	0.783	0.815	0.727	0.857	0.667
<code>open_closed_mismatch</code>	12	0.556	0.286	0.632	0.400	0.471	0.400
<code>agree_disagree_scale</code>	13	0.526	0.000	0.435	0.111	0.615	0.435
<code>unbalanced_scale</code>	12	0.250	0.000	0.435	0.000	0.400	0.000
<code>incomplete_options</code>	12	0.182	0.353	0.174	0.333	0.213	0.571
<code>non_exclusive_options</code>	12	0.545	0.000	0.200	0.000	0.316	0.000
<code>missing_scale_labels</code>	20	0.811	0.000	0.800	0.000	0.872	0.261
<code>too_many_scale_points</code>	12	0.960	0.909	0.880	0.909	0.774	1.000
<code>polarity_mismatch</code>	12	0.786	0.762	0.667	0.700	0.759	0.783

A second recurring pattern was systematic baseline overprediction of `incomplete_options` and `loaded_question`. Under P0 baseline, these labels generated 78 and 56 false positives, respectively; under P2 they still generated 72 and 43. This produced high recall for the labels but very low precision. Orchestration sharply reduced these false positives, but often did so by suppressing the label entirely rather than resolving the boundary reliably. The improvement from P2 orchestration is notable because it increased supported predictions while retaining this restraint.

E Supplementary Revision Results

Oracle-revision results are shown in Table 14. For Qwen, P2 increased exact-option matching from 19 to 23 items in the baseline and from 22 to 23 under orchestration. Question exact match changed little, and SARI did not improve consistently. The largest exact-full rate in any complete non-thinking oracle run was only 0.025 (4 of 160 items) for Qwen P1 orchestration.

Table 14: Oracle-revision metrics without thinking. Gold categories are supplied, but item-specific evidence and repair suggestions are not.

Model	Prompt	Architecture	BERTScore	SARI	Exact Q.	Exact options	Exact full	Changed	Scored
Mistral-7B	P0	Baseline	0.952	50.95	0.075	0.100	0.019	0.850	160/160
Mistral-7B	P0	Orchestrated	0.951	58.17	0.169	0.110	0.013	1.000	154/160
Gemma-2-9B	P0	Baseline	0.953	58.30	0.087	0.100	0.013	1.000	160/160
Gemma-2-9B	P0	Orchestrated	0.955	61.70	0.180	0.107	0.013	1.000	150/160
Qwen3.5-9B	P0	Baseline	0.956	62.46	0.181	0.119	0.019	0.950	160/160
Qwen3.5-9B	P0	Orchestrated	0.956	61.94	0.194	0.138	0.019	0.950	160/160
Qwen3.5-9B	P1	Baseline	0.954	61.45	0.175	0.119	0.013	0.963	160/160
Qwen3.5-9B	P1	Orchestrated	0.956	62.02	0.188	0.138	0.025	0.944	160/160
Qwen3.5-9B	P2	Baseline	0.955	61.18	0.181	0.144	0.019	0.825	160/160
Qwen3.5-9B	P2	Orchestrated	0.956	61.61	0.194	0.144	0.019	0.906	160/160

Oracle revision was therefore not a strict empirical upper bound. P2 baseline left 28 gold-flawed items unchanged and P2 orchestration left 15 unchanged, despite receiving the correct categories. Inspection of the revision notes showed that the revisers sometimes rejected a gold label because the oracle issue object contained `evidence=null`. For example, some `loaded_question` items were preserved because a premise-denial option was interpreted as sufficient evidence that no stem repair was needed. This behavior follows the prompts’ conservative evidence gate, but it means that oracle mode measures revision from correct label-only guidance rather than revision from a fully specified expert diagnosis. It also explains why oracle scores did not consistently exceed end-to-end scores. The 40 clean controls are preserved unchanged by construction in oracle mode and are excluded from the revision averages; their perfect preservation is not treated as model performance.

F Supplementary Thinking-Mode Results

F.1 Completion Reliability

Table 15 separates items reached, successful structured outputs, failed outputs, and unprocessed rows. The end-to-end baseline produced only 57 successful outputs and the orchestrated version 62. The baseline oracle run reached all items but failed on 56; the orchestrated oracle run stopped after 128 items. Token exhaustion was the main operational cause: the model often consumed the generation budget on internal reasoning and then returned truncated, empty, or schema-invalid JSON.

Table 15: Thinking-mode completion. The final column gives successful clean, single-label, and multi-label items.

Architecture	Mode	Reached	Successful	Failed	Unprocessed	Successful C/S/M
Baseline	End-to-end	149	57	92	51	21/33/3
Baseline	Oracle	200	144	56	0	40/74/30
Orchestrated	End-to-end	138	62	76	62	19/42/1
Orchestrated	Oracle	128	92	36	72	40/52/0

F.2 Conditional Revision Result

Only the baseline oracle thinking run produced final semantic revision metrics. Table 16 reports them conditionally on the 104 successful flawed items. Exact-option matching reached 22 of 104 items (0.212), but no comparison is made with a full non-thinking run because that would use a different item set. A matched-subset semantic rescore was not contained in the archived artifacts.

Table 16: Conditional revision quality for the completed P2 baseline-oracle thinking outputs.

Scored	BERTScore	SARI	Exact Q.	Exact options	Exact full	Changed
104/160	0.958	63.47	0.154	0.212	0.048	1.000

G Exploratory Semantic Option Matching Metric

This appendix outlines a proposed, exploratory metric that was not used in the experiments reported in this paper.

A further direction concerns evaluation of response-option revisions. Exact option matching is intentionally transparent but very strict. Semantically equivalent revisions such as “Very dissatisfied” and “Very unhappy” receive no credit unless their normalized strings are identical. An exploratory *Semantic Option Matching F1* (SOM-F1) could address this limitation while still penalizing missing and additional options. For gold option set E and predicted option set P , a semantic model would score all predicted–gold option pairs and optimal one-to-one matching would be obtained with the Hungarian algorithm. If M denotes the resulting matching and $s(e, p)$ the semantic similarity of a matched pair, the metric would be

$$P_{\text{sem}} = \frac{\sum_{(e,p) \in M} s(e, p)}{|P|}, \quad R_{\text{sem}} = \frac{\sum_{(e,p) \in M} s(e, p)}{|E|},$$

$$\text{SOM-F1} = \frac{2P_{\text{sem}}R_{\text{sem}}}{P_{\text{sem}} + R_{\text{sem}}}.$$

A defensible implementation would use a pretrained cross-encoder, preferably one trained for natural-language inference. Bidirectional entailment could provide evidence that two options express approximately the same response meaning, while high contradiction probability could prevent opposite-polarity options such as “Very satisfied” and “Very dissatisfied” from receiving misleadingly high similarity.

Before SOM-F1 could be used as a research metric, it would require an independent human-labelled validation set containing synonymous options, opposite polarities, neutral categories, frequency and quantity scales, numeric ranges, residual options, and missing or additional categories. Model selection and calibration should be performed only on development pairs, followed by evaluation on held-out pairs.